

Problem 2

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Problem 1 Evaluate the following definite integrals:

(a) $\int_{-2}^1 t^2 \sin(t^3) \, dt$

(b) $\int_0^{1/2} \frac{13e^u}{3e^u - 5} \, du$

(c) $\int_0^{\pi/2} \frac{\cos(x)}{4 + \sin(x)} \, dx$

(d) $\int_1^4 \frac{e^{\sqrt{x}}}{3\sqrt{x}} \, dx$

(e) $\int_{\pi/3}^{\pi/2} \sin(x) \sec^2(\cos(x)) \, dx$